

# System Design

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## AI Evaluation Platform

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### Overview

The AI Evaluation Platform is a Retrieval-Augmented Generation (RAG) based system that evaluates AI-generated answers using a reference knowledge base. The system accepts a question and an AI-generated response, retrieves the most relevant context from a vector database, evaluates the response across multiple quality dimensions, and returns a final verdict.

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## High-Level Architecture

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User

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Streamlit Frontend

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FastAPI Backend

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Evaluation Orchestrator

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RAG Retrieval Module

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ChromaDB Vector Database

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Sentence Transformer Embedding Model

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Knowledge Base (SQuAD + TruthfulQA)

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## Modules

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### 1. Evaluation Input Module

#### Responsibilities:

- Accept question
- Accept AI-generated response
- Accept optional reference answer
- Validate user input
- Send evaluation request to backend

#### Technology:

- Streamlit
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## 2. Reference Knowledge Base

#### Responsibilities:

- Store benchmark datasets
- Store document chunks
- Generate embeddings
- Maintain vector database

#### Datasets:

- SQuAD
- TruthfulQA

#### Embedding Model:

- all-MiniLM-L6-v2

#### Vector Store:

- ChromaDB
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## 3. Retrieval Module

#### Responsibilities:

- Convert user question into embedding
  - Search vector database
  - Retrieve Top-K relevant chunks
  - Return retrieved context
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## 4. Evaluation Module

#### The evaluation module computes:

- Relevance
- Accuracy

- Hallucination
- Completeness
- Overall Score

It also generates the final verdict.

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## 5. Report Generation

Responsibilities:

- Generate PDF evaluation report
  - Store evaluation summary
  - Export evaluation results
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## Orchestration Flow

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User Input

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FastAPI API

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Retrieve Context

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Evaluate Response

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Generate Scores

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Generate Verdict

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Return Results

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Display Dashboard

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## Data Flow

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Question

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Embedding

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Vector Search

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Retrieved Context

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Score Calculation

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Final Verdict

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User Interface

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## Design Goals

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- Modular architecture
- Easy to extend
- Scalable
- Retrieval-based evaluation
- Explainable scoring
- Simple deployment